Puma Marketing Strategies and Parkruns

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Overview

Key Insights

Question One: Which Parkrun provides the best marketing opportunity with the largest number of people?

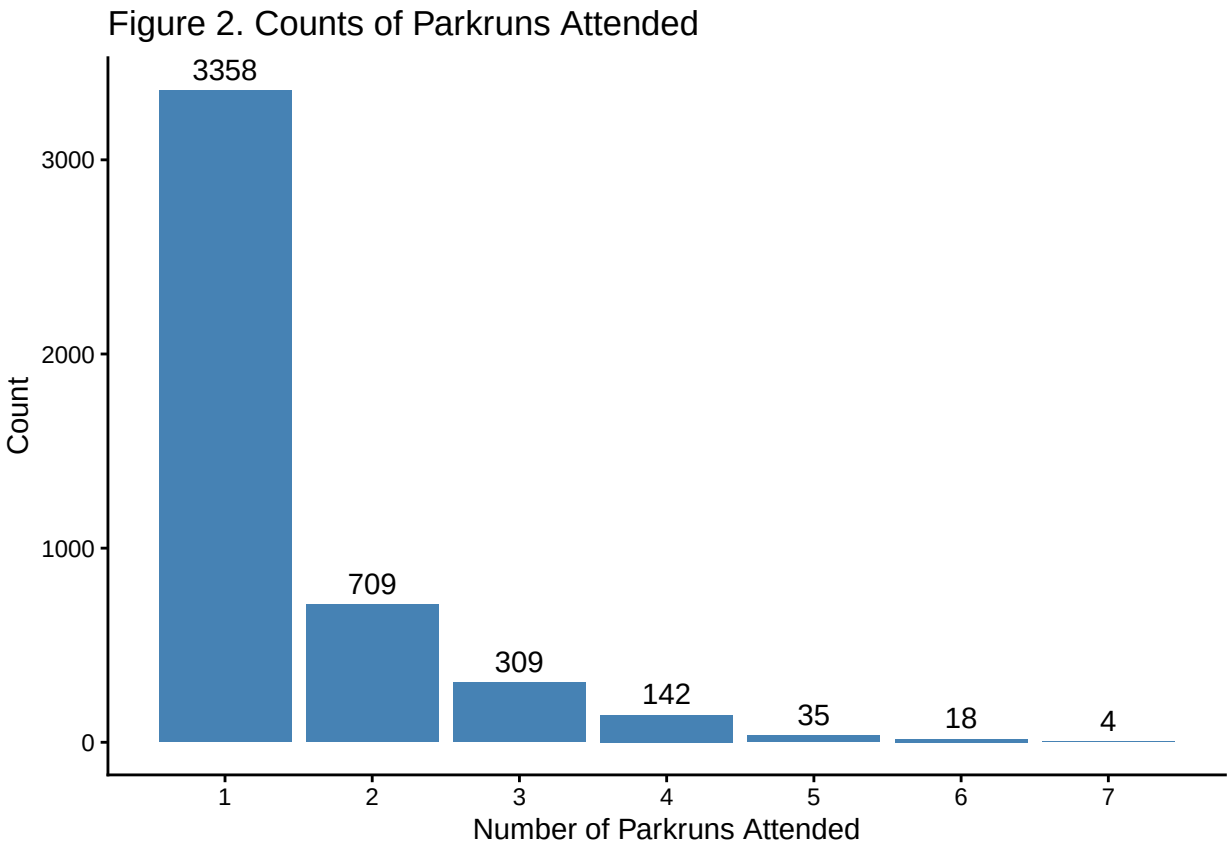
Question Two: Do new people come each week or is it always the same people?

Question Three: Do people move between Parkruns? How does this affect advertising?

In order to determine whether or not runners attend different Parkruns, we can simply count the number of Parkrun locations each runner has been to in the past 11 weeks. However, due to unidentifiability, we cannot include any unknown runners in this question.

Table 1: Proportion of runners who have attended a different parkrun

Total Runners	Mixed Runners	Percentage of Mixed Runners
4575	1217	26.6



The above table (Figure 1) shows that there is a significant number of runners who have attended Parkruns at mulitple locations (26.6%), or about 1200 runners out of 4600. However, Figure 2 shows that the vast majority of runners have only attended events at one Parkrun location, with very few reaching out to others.

However, this 26% overlap means that advertising across all locations is not going to be particularly budget effective, as advertisements will end up being displayed to people who have already bought Puma products. Therefore, advertising across fewer Parkrun locations can be a more economical option, as it would allow for the same effective advertisement reach, at a lower cost when compared to advertising at all locations.

Question Four: If Puma want to target elite runners, where should they focus their efforts?

In order to target elite runner for the marketing campaign, we can look at the "Age grade" of each runners.

"Age grade" is a percentage that compares runner's time to the world record for runner's age and gender.

This percentage could be understood as: - <50% = beginner - 50-60% = average - 60-70% = good - 70-80% = very good - 80%+ = excellent

A person who has an age grade above 70 would be consider as an elite runner, this applies to both men and women. Here is the table shows the number of elite runners between parkruns.

```
# Overall elite analysis by location
elite_analysis_overall <- clean_data %>%
  group_by(parkrun) %>%
  summarise(
    total_runners = n(),
```

```
    elite_runners = sum(elite_runner),
    elite_percentage = elite_runners / total_runners * 100,
    avg_grade = mean(grade, na.rm = TRUE)
  ) %>%
  arrange(desc(elite_runners))%>%
  rename(
    "Parkrun Location" = parkrun,
    "Total Runners" = total_runners,
    "Elite runners" = elite_runners,
    "Elite runners percentage" =  elite_percentage,
    "Average grade" = avg_grade
  )

elite_analysis_overall %>%
  kable(
    caption = "Table 4.1. Overal Elite Analysis by Parkrun Location",
    digits = 2
  )
```

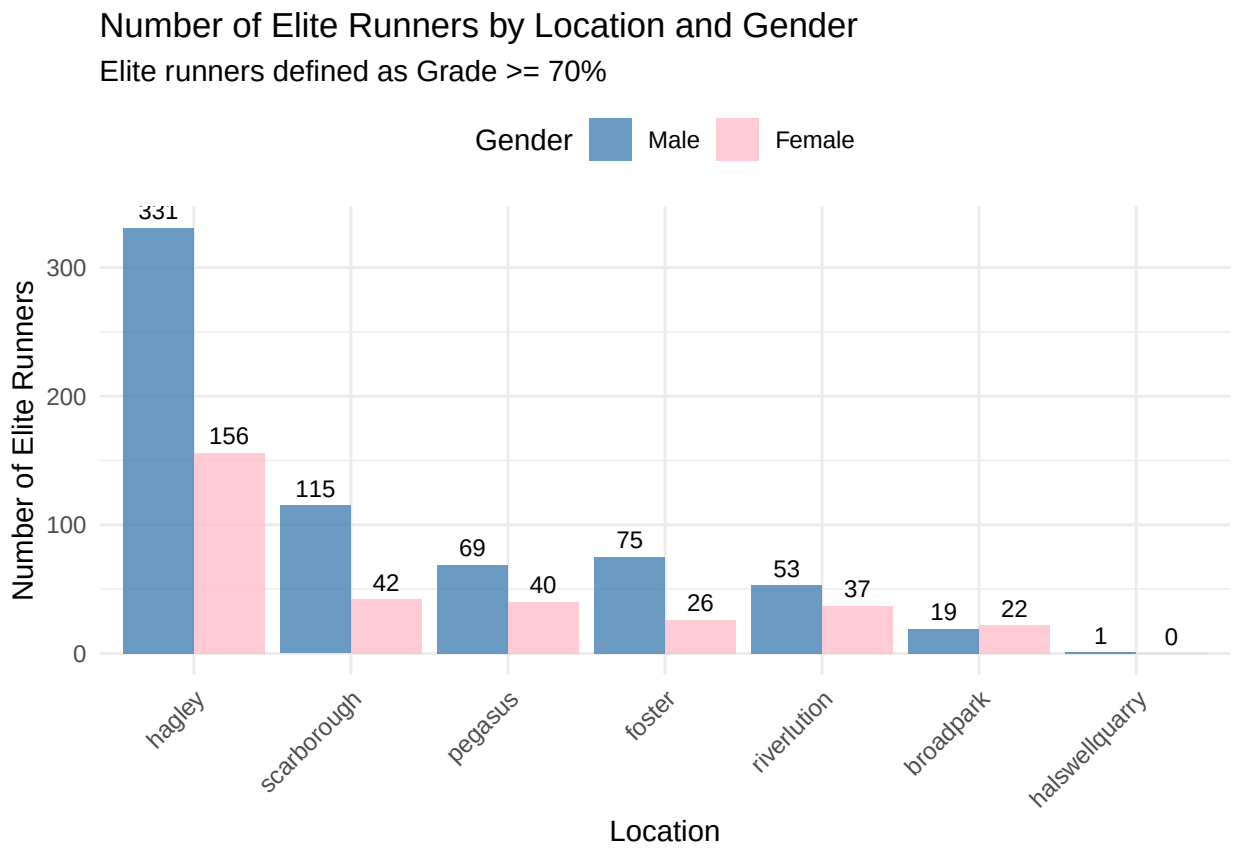
Table 2: Table 4.1. Overal Elite Analysis by Parkrun Location

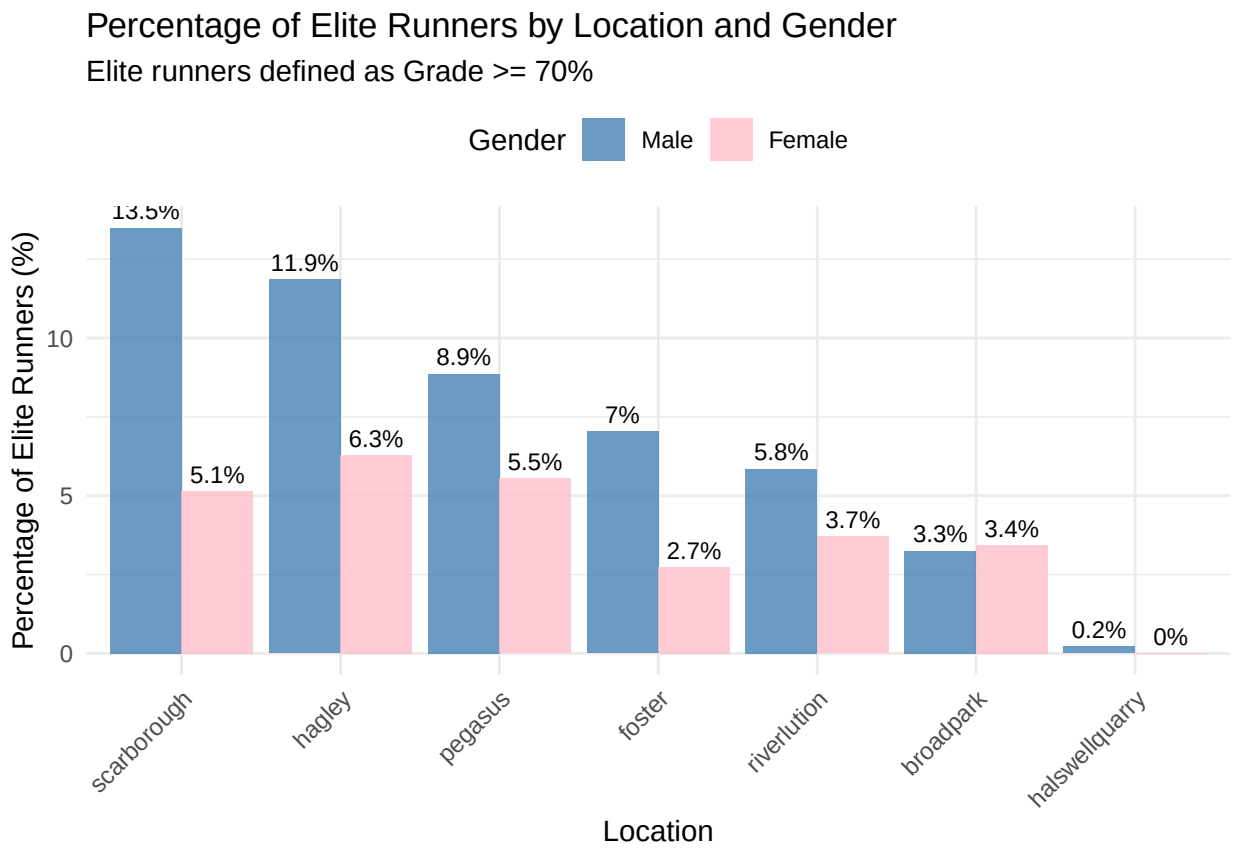
Parkrun Location	Total Runners	Elite runners	Elite runners percentage	Average grade
hagley	5274	487	9.23	53.02
scarborough	1669	157	9.41	53.71
pegasus	1500	109	7.27	52.24
foster	2018	101	5.00	50.45
riverlution	1904	90	4.73	49.88
broadpark	1225	41	3.35	48.81
halswellquarry	769	1	0.13	43.18

It's noticeable to find that the Hagley parkrun has an outstanding numbers of elite runner but Scarborough parkrun has the highest elite runner percentage among these parkruns.

Table 3: Table 4.2 Elite Analysis by Gender and Parkrun Location

Parkrun Location	Gender	Total Runners	Elite runners	Elite runners percentage	Average grade	Max grade
broadpark	M	584	19	3.25	49.74	80.28
broadpark	F	641	22	3.43	47.96	84.31
foster	M	1066	75	7.04	53.86	88.35
foster	F	952	26	2.73	46.63	86.23
hagley	M	2792	331	11.86	54.97	90.27
hagley	F	2482	156	6.29	50.83	92.03
halswellquarry	M	457	1	0.22	45.33	72.54
halswellquarry	F	312	0	0.00	40.02	63.66
pegasus	M	779	69	8.86	53.51	81.87
pegasus	F	721	40	5.55	50.88	85.84
riverlution	M	906	53	5.85	51.72	84.59
riverlution	F	998	37	3.71	48.22	85.74
scarborough	M	852	115	13.50	56.14	85.51
scarborough	F	817	42	5.14	51.18	85.58


Number of Elite Runners by Location and Gender
Elite runners defined as Grade >= 70%
Gender
Male
Female
Number of Elite Runners
0
100
200
300
331
156
115
42
69
40
75
26
53
37
19
22
1
0
hagley
scarborough
pegasus
foster
riverlution
broadpark
halswellquarry
Location

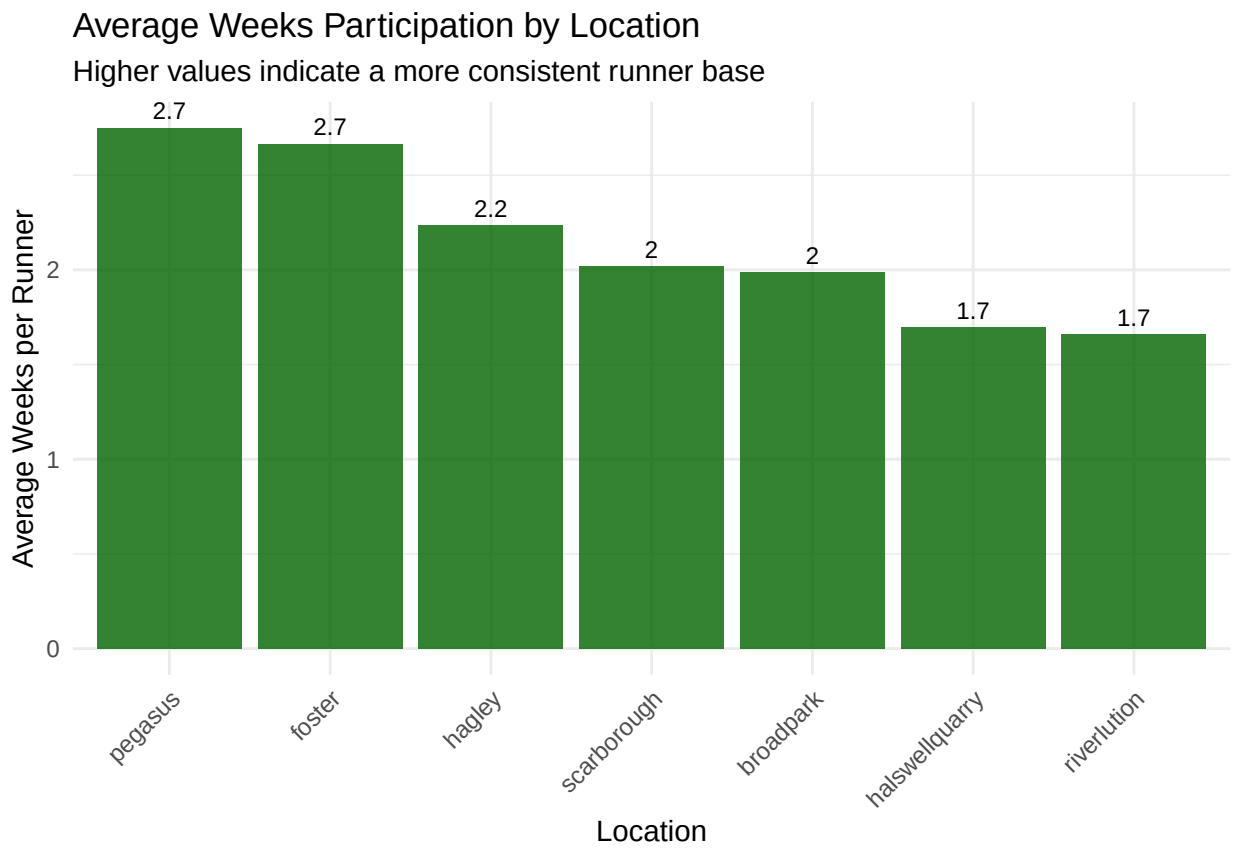


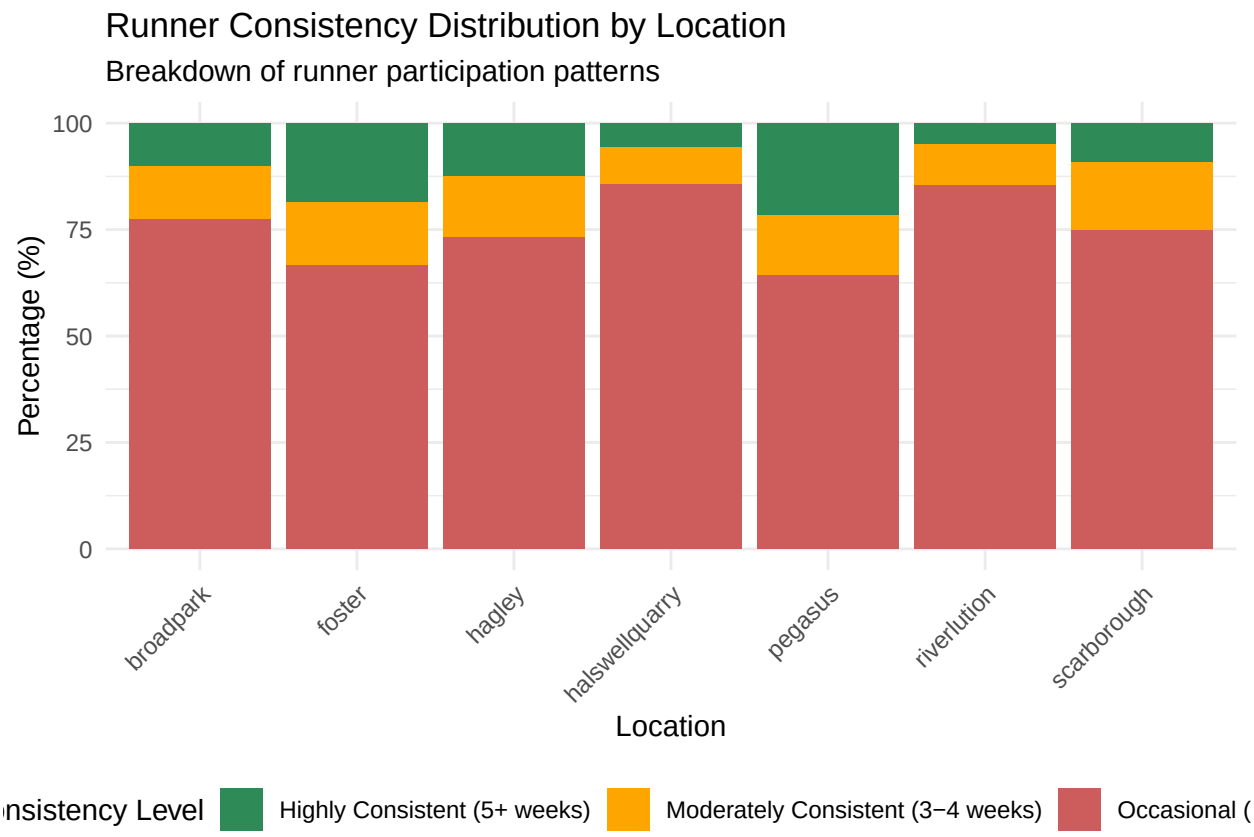
Question Five: Puma believe more consistent runners have better potential as customers, how many runners are consistent and is consistency different across locations?

table consistent

Table 4: Table 4.1 Number of consistent runner in each parkrun

Parkrun Location	Total Unique Runners	Highly Consistent (5+ Weeks)	Moderately Consistent (3–4 Weeks)	Occasional (1–2 Weeks)	Percentage Highly Consistent (%)	Average Weeks per Runner
pegasus	539	116	76	347	21.52	2.75
foster	745	138	111	496	18.52	2.66
hagley	2300	285	330	1685	12.39	2.24
scarborough	809	75	128	606	9.27	2.02
broadpark	603	61	75	467	10.12	1.99
halswellquarry	449	25	39	385	5.57	1.70
riverlution	1131	56	110	965	4.95	1.66


Average Weeks Participation by Location
Higher values indicate a more consistent runner base
Average Weeks per Runner
0
1
2
2.7
2.7
2.2
2
2
1.7
1.7
pegasus
foster
hagley
scarborough
broadpark
halswellquarry
riverlution
Location



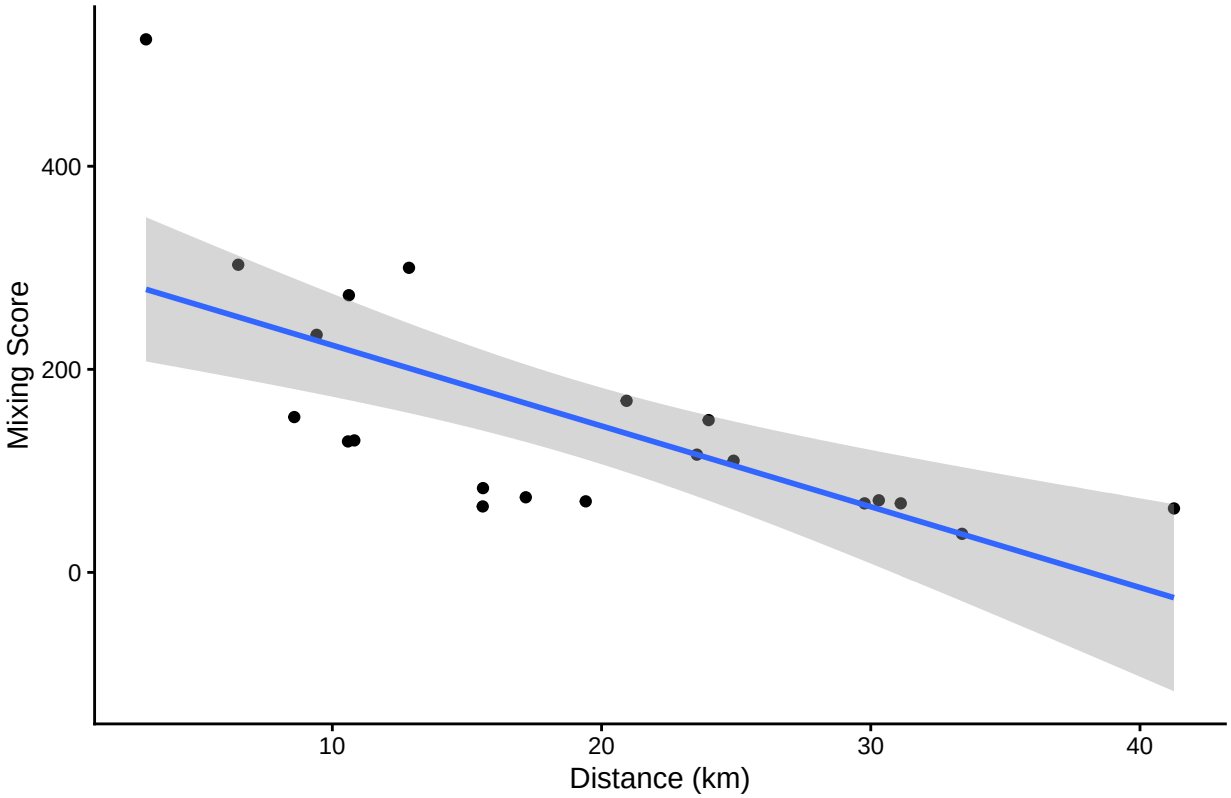
Question Six: Does distance between Parkruns affect how much mixing occurs?

In order to determine whether or not the distance between two Parkruns predicts how much mixing occurs, we can obtain each unique pair of Parkruns, and then using geographical data, determine the distances between these pairs, effectively creating a map of each Parkrun location.

We can then count the number of runners who have gone to both locations in these pairs, and set that as our Mixing Score. A higher Mixing Score means more runners have attended both Parkruns in the past 11 weeks.

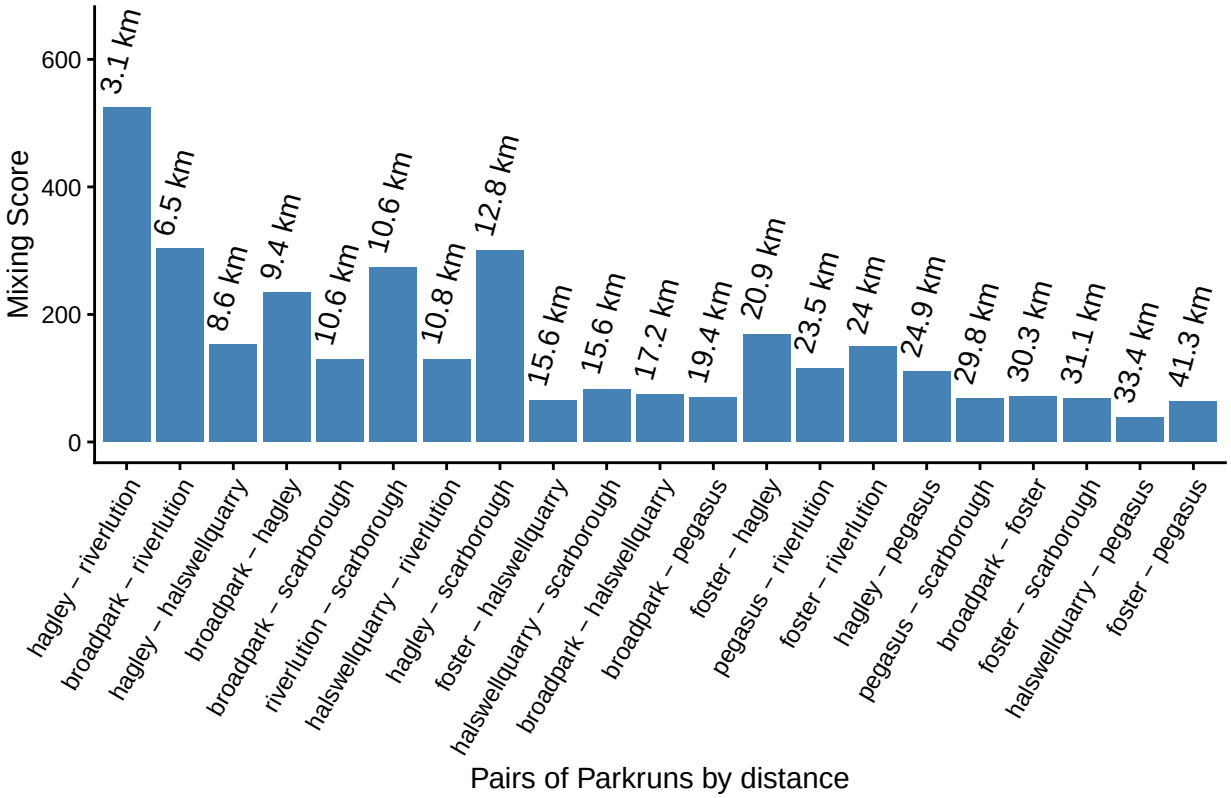
```
## `geom_smooth()` using formula = 'y ~ x'
```

Figure 3. Mixing Score vs. Distance



As seen above, Figure 3 shows that there is a negative correlation between Distance and Mixing Score. In other words, as the distance between parkruns increases, the amount of mixing decreases.

Figure 4. Parkrun Pairs and Mixing Scores arranged by distance



The above bar chart (Figure 4), also shows this correlation, but also labels each parkrun pair with the distance between them. As is seen in the chart, the Hagley and Riverlution parkruns are very close (only 3km apart), and as such, receive a very large amount of mixing.

As for how this affects advertising, this allows for more control over audience overlap. Given the amount of mixing between the Hagley and Riverlution parkruns, advertising at just one of these parkruns would minimise the amount of overlap and spending, without reducing the effectiveness of the advertisements. However, given the amount of distance between the Foster and Pegasus parkruns, advertising at only one location would only harm any marketing efforts.

As such, for pairs of parkruns with high amounts of mixing, advertising at just one location would be the best option, preferably the parkrun which is generally more popular/has the most runners.

Summary and Conclusion